Goodfriend Memorial Lecture: Automation, Learning and Career Dynamics

By Erik Hurst

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Introduced in 2023, the Goodfriend Memorial Lecture series honors the legacy of Marvin Goodfriend, long-time Richmond Fed economist, research director and senior policy advisor. The lecture is delivered as part of the Richmond Fed's Collaboration of Research Economists (CORE) Week model, which brings together Richmond Fed economists and visiting economists from a range of disciplines for seminars, conferences, networking and collaboration.

On Aug. 6, 2026, Erik Hurst delivered the Marvin Goodfriend Memorial Lecture titled "Automation, Learning and Career Dynamics." Hurst is the Roman Family Distinguished Service Professor of Economics and John E. Jeuck Faculty Fellow at the Booth School of Business at the University of Chicago. He is also the director of the Becker Friedman Institute for Economics.

The following is a lightly edited transcript of the lecture.



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We're going to do this as a regular seminar, so anytime you have questions, just come through as opposed to a regular fancy plenary thing.

Claudia [Macaluso] asked me to give this paper to talk a bit about the AI theme that is going on. I have to say, first approximation as an applied theorist, I'm inherently uninterested in AI. What I mean by that is that, for many things, it's just like any other technology. Now it might be large, and it might be touching different types of workers than before. But in terms of theory, it looks very similar to other technologies we've done before, maybe with the exception of two different things. One of which is stuff Martin [Beraja] has worked on before. It might be happening fast, and a lot of our theories don't really integrate the speed at which things are happening. That's theoretically interesting to me, but Martin's already done that.

It is in some ways about how it interacts with human capital, and that is something else I've been trying to think about, because this is something that might substitute for human capital. And that might have enough incentives to accumulate — or decumulate — human capital over time. That's where this paper comes from.

There's been a lot of industry-like people talking about the role of AI and its potential interaction in human capital. Some are very dire. There are quotes I put up here from the CEO of Anthropic that AI could wipe out half of entry-level white-collar jobs in the next five years. That's been walked back in recent periods of time. But the idea that AI could substitute for workers in entry-level positions on the early rungs of a job ladder is out in the ether. When you start taking a look at some of these consulting firms, they are worried

about how this could affect the quality of managers tomorrow: If we don't have any young people coming into our firms today to learn about how the organizations are run and how to build institutional knowledge going forward, who will be leading companies tomorrow?

It's very similar to our PhD students: If we use AI and we're not letting them work with us, teach with us, grind like we used to grind in the old days, is the idea going to be as good tomorrow as they would have been today? We'll call that the pessimistic view that's out there.

Conversely, there's a very optimistic view out there that AI is going to be so wonderful. Martin's was talking about this paper — that it's going to make us learn so much faster going forward. So it's going to, in one sense, expand the amount of ideas we can create. It is then going to be a complement for workers, particularly at the higher rungs of the job ladder who do the innovation at the top rungs. And maybe the AI itself, it will have more things to learn and will maybe accelerate the learning process, so that the human capital pipeline is going to be maintained.

With that as background, let's move to the theory. This is a pure applied theory paper. As I told Boyan [Jovanovic] earlier, our model was something Boyan would have written in 1988. That's the kind of model we're having for this paper.

Here's what we're going to do: We're going to ask how a technology — call it AI — affects human capital accumulation with endogenous learning on the job. We'll try to nest both the pessimistic and optimistic views by having a model with a continuum of tasks that differ in their amenability to AI automation. Just like in Acemoglu-Restrepo, there will be some substitutes between AI and certain types of people on the job ladder.

We're going to have AI also affect the ability of people at the upper rungs of the job ladder — we'll call them managers — to create new ideas. So there's a frontier task choice in this model, and AI could complement that. And we're going to have a job ladder, or call it a career ladder. We're going to have entry-level workers who execute tasks, and we're going to have higher-level rung workers who are going to create new tasks. There's going to be learning along the ladder, and there's going to be endogenous career choice. Worker human capital is going to depend on the amount of stuff you do — learning by doing at the individual level along the ladder. And then we could talk about the fact that learning more along the way makes us more productive as managers and better able to use this AI technology when it comes along.

That's going to be the framework, and now we're going to work through and show you what this model is going to do. I'm just going to tell you two things: This model is going to generate multiple steady states; there is going to be both high-learning and low-learning environments.

Audience question: Erik, before you go there, can I ask you: Do you think at all about bundling the association of different tasks into a role and therefore into a human? Whether that's changing at all?

Let me show you the model. The part that's going to be new is really the career ladder. There's going to be a worker who's going to do some stuff. There's going to be machines that are going to do some stuff. The amount of stuff machines do relative to workers is going to be changing. Workers are going to accumulate human capital as they go along. Eventually they become managers, and that human capital makes them more productive.

Once I show you the model, it's not going to have these nice bundles, but it'll be in one type of sector. Now once we lay over top of it all the other traditional parts of technological change and labor market stuff we've been doing for decades, some sectors will be more amenable to AI, others are not. Workers can substitute across sectors. All of that, we know how to deal with. So what we're adding is a layer on top of it.

I'm going to walk through how we could get these multiple steady states, but a low-learning environment means low human capital. Labor is going to be relatively expensive in the human capital space because the supply of human capital is going to be low, which is then going to lead to lower-quality managers. Expensive labor leads to more crowd-out, which leads to less innovation, which then leads to fewer things for people to learn about, which generates the low-learning environment. Conversely, the high-learning environment does the exact opposite.

This simple model is going to have these rich multiple steady states. But what I want to lead to at the end is that when you put these pieces together in a standard model, the optimistic view tends to be rather robust through many of the model specifications. I'll tell you what I mean precisely once we get into the model, but after doing this, it's much more robust.

The fact that the task frontier could be expanding at the same time, even when I don't allow Martin's effect that the technology could also make us learn faster, that's also going to amplify all this. That's going to have a force until AI gets really, really cheap, we're still going to be expanding the task frontier, and you still need people to do that, and that human capital is still going to be there.

Once AI does get very cheap, that's Martin's story: We all become "kings" at that point. AI is going to make the innovation for us. AI is going to do the work for us. We're going to be in a world where there's going to be low human capital. We all could be stupid because there's nothing valuable for us to do because the machines are going to create the value going forward. And in that kind of world, that's at the end, but until we get to that point, there's still a role for human capital and forces that make that sustained. I want to show you as we go through some of that. Then we'll talk about what happens when there's both a planner in this economy and maybe there's some inefficiency. We'll talk about that as we go along. Okay, that's my preamble.

Let's get into the model. The model is going to have some ingredients: Time is continuous. There's a unit measure of individuals and a representative but atomistic firm in the background. Workers differ only in a human capital sense, which is accumulated over their lifetime. We're all born exactly the same, then we accumulate human capital over our lives. It's a perpetual-youth model: We die off at some rate, χ, and we're born with the minimum level of human capital, which I'm going to renormalize to 1. We could talk about what happens if there's growth over time, but you're just born with the minimum level of human capital.

Audience question: I was going to ask the question at the other end. If you had a finite or accelerated death rate — so as you reach a certain age, you'd have less incentive to accumulate — it would still … that's just a quantitative matter?

That's just a quantitative matter. You could change how fast we're dying or not, because it otherwise is perpetual youth. We're living forever, and we're just dying off. If you want us to die off a little faster to shorten the horizon, that's just going to be a χ going …

So the individuals are going to accumulate human capital by learning-by-doing on the job. If machines do it, I learn less; if I do it, I learn more. The individual has one decision in this model: when to transition to being a manager. When do they start innovating?

The firm is going to do a bunch of things. They're going to hire workers, hire managers, create new tasks and choose whether to automate existing tasks.

On the production side of the economy, it's going to be very Acemoglu-Restrepo, with one little tweak. The tweak is going to be the amount of tasks in the economy are going to be endogenous. We'll call that F, the task frontier. That's going to be something that's going to be endogenous. I'll tell you the firm decision that's going to drive that on the next slide.

Conditional on that, firms get output of different types of tasks that are going to be done in the economy, and there's going to be some diminishing returns in the tasks. When you're producing at the task level, workers are perfect substitutes for machines, just like in Acemoglu-Restrepo. So firms could produce with some workers. Those workers are going to have some human capital, and they could integrate over the amount of human capital that workers have when they're hiring workers for every given level of task. Or you could produce that task by buying some technology. And that technology, in terms of the task production, has some productivity z. We order the z's in such a way so they're decreasing: This is just ordering that when you start automating, you start automating the low z first, and then eventually the high z's are the last to get automated. This is going to generate just like Acemoglu — eventually I'm going to show you first-order conditions and optimizations later — there's going to be a task cutoff, below which you're going to use machines and above which you're going to produce with workers. The technology is going to be able to be purchased at some exogenous price Q — that's going to be like the AI in the model. AI is going to have a price Q. You could choose to buy it in terms of your use of automation.

Audience question: Erik? This goes to Claudia's question. I'm not quite sure I see the relationship between tasks and people.

In this model, you're a consultant at a consulting firm, and you're doing a lot of stuff: You're making presentations; you're sitting in on client calls; you're doing some analysis for a firm.

Audience question: Stop! That sounds like hell. {Laughter} I'm looking up here, so you've got an x. You're integrating over x, right? And you've got τ. You're not integrating over τ.

So for x at any given level, my x could be applied to any of the tasks in some optimal way by the firm. For anything that I'm doing, I have some x, and I'm going to be allocated to a bunch of tasks because that's the only heterogeneity in the model.

Audience question: So each task could have all these different *x*. You don't have *x* indexing anything, right?

Right, it's just our "name." Our name is *x*, and so over time, I'm born, and then my *x* is evolving over time. Think of that as the human capital level of the workforce. Firms use the human capital of the workforce to produce the amount of stuff that's not done by machines. Some of us are allocated to this. There's no real matching of *x* to any other task. It's all perfect substitutes at that level. And the way you should again think about it, is as a career, you're moving along, and then your stuff gets automated at the consulting firm (or law firm, or data science place), and then you're doing the rest of the tasks available. By doing those things, you're going to learn some stuff. The whole paper is about what you learn. This is all kind of standard stuff so far. Once we get into it, it's going to be the learning function and how we're going to specify that.

Audience question: Does it matter that all tasks can be done by either human capital or technology, and that you do not have tasks that are only human or only machine?

It's a continuum, with cutoffs where this thing could be zero, this thing could be 1. And if this thing is one, everything's done by humans; if it's zero, none are done by humans.

Audience question: But in principle, you could have all those? There is not a technology constraint that says, no matter how much x you have, you can't possibly do this?

The whole *z* function could have some things that you're always doing. We could make the *z* function anything we want to make the shape as steep or as flat at different parts. I'm going to specify a functional form for that *z* function. That is part one.

Part two adds Acemoglu-Restrepo with an endogenous task frontier. There are going to be different types of labor— we'll call it the human capital of managers, *h(x)*. So it's *x* plus a little extra when you become a manager. Maybe you get a little extra along the way. And

people endogenously choose when to be a worker and how much to be a manager, and that's going to be your human capital. You're going to integrate the human capital stock of people who choose to be managers. That human capital stock of the managers could be a complement with some other technology, also priced at Q. We'll call it me using Claude in my office as opposed to my graduate students to create new ideas. So you take Claude, you take me, there's some elasticity of substitution between Claude and me in creating new ideas.

Audience question: Erik, you have perfect substitutes in production and perfect substitutes here. The story is important.

It's going to be the relative substitutability of workers relative to the relative substitutability of managers. Exactly, and that's going to be the key right now. It's going to have imperfect substitutability between managers and perfect substitutability between the workers. What's really important is the relative substitutability between the two.

Audience question: This third ingredient of yours — that if you had high human capital, you become a better manager — that's testable in principle. It's not obvious to me that that would be the case. You might get promoted when you're obsolescent.

We'll talk about data as we go along, but there are learning-by-doing models out there suggesting that as you go through your career, you become more productive. We're saying the people who want to become managers have some skills that aren't the same as the entry-level people. That's the first part. And the second part is the people who become managers are more likely to create some ideas, at least within the firm. If you look within the firm, a lot of the creation of new tasks is done by upper management. The execution of tasks is done by others.

It's not the same kind of Haltiwanger model where we could have innovative new people on the outside. I'm thinking about within the firm as we're going through, and there are some people who make new product-line decisions and people who are executing those. We're saying that sometimes when you execute, it can make you better at making the things.

The fact that wages slope over the life cycle upside is a component of the data that is something that moves toward the learning-by-doing model of reasons why people accumulate human capital over the job.

There's one other thing in the background: Maybe I can't learn unboundedly. There's only a certain amount to learn, so we have a frontier amount of knowledge in the background. That's going to keep the equilibrium nice and bounded for us to think about. There's a whole bunch of other parameters I'm not going to talk about today, like the weights on how much is technology versus people in the production function — all of that is relatively standard.

This is just a standard occupational choice model on top of this. On the next slide is where the all the economics is going to be, or at least the deep part of it. How are we going to learn? Right now, we're just saying the choice of the worker is choosing when to become a manager or when to stay a worker. If they stay a worker, they get the wage of the worker times their human capital. If they become a manager, they get the wage of the manager plus a bit of extra human capital. Maybe managers can use their human capital slightly better, or maybe not, we could set that equal to 1. And you're going to choose based on the wages you're getting and any option value of staying in your career choice going forward.

The key thing, which I'll define formally on the next slide, is this φ is going to be the learning that's going to be in the model. If you stay a worker, you learn a bit more going forward, which makes you more productive as a manager. Managers can also keep learning as managers.

So what is the learning? In this model, your human capital growth is a function of the level of your human capital and this φ term. φ is going to be the rate at which you learn, and we assume this has two components. First, some parameter Δ, which could or could not move with technology. (Today, I'm assuming it doesn't move with technology.) In Martin's paper, maybe we learn a little faster on the worker side, not the firm side, with new technology that comes along. I'm going to shut that down. That's going to do in the model the same thing that frontier expansion does: It would make people learn a lot more. And what is the learning? You're going to learn by the amount of tasks done by workers. So I'm in the firm, and the firm automates τ* tasks. There are F tasks available to be done; F – τ* is the part done by me, and I'm going to do a little piece of all of those, and as a result, I learn along the way from the stuff that workers are doing, as opposed to the firm. That's the learning-by-doing mechanism.

Audience question: This φ — this is learning. This human capital also affects me when I'm a manager, right?

Yes, exactly. It's going to be an option value.

Audience question: But if the whole firm is a machine or an AI, why does it matter if I learn the tasks enough to be a manager, in that case?

That's an equilibrium in the model. When the price of technology gets sufficiently low, we don't even need managers. So there, the value of human capital drops for everyone. That's an equilibrium in the model as well. I'm going to show you, that's going to be the level of the price of technology relative to the marginal products of other stuff. But if it gets sufficiently low, that's going to be an equilibrium in the model.

Audience question: Erik, can I ask something here? In the model, can everyone become a manager if they want to?

Yep. In equilibrium, day one, everybody becomes a manager.

Audience question: What if the firm chooses who can be a manager, depending on your x?

We've thought about this. An equilibrium might be that we all become managers and we're all "stupid": None of us learns anything, and we're all creating a little product. That could be an equilibrium. Now, if there's a tournament model: Suppose we only need three managers going forward. I haven't written this model, but it's one I've been playing with in my mind: If the firm can observe who's good the more stuff we do, then we could learn a little bit more. I think you'd get a very similar implication if the firm was choosing. It was bounded, and they had to choose which one of us is good, and they can only tell which one of us is good by us doing stuff. It'll have a very similar flavor.

Audience question: So the strategy would be: When your x is high enough, I promote you to manager. That creates an incentive to accumulate human capital, right?

It's the same accumulation here, because the worker's choosing to be more productive when they're a manager. I don't think it matters much whether the firm or the worker makes the choice. I don't think that's going to be much of a distinction here. The question is if there's a capacity constraint that the firm could put on. That's the one I was thinking about. Otherwise, the firm and the worker will make the same choice based on marginal products. If your marginal product is sufficiently high in this type of activity, you're going to do that activity, and the workers and the firms will both internalize that.

Audience question: So we learn in proportion to the fraction of tasks that aren't automated. It's not about how many people are actually workers versus not. In your motivation, it seems like some people are going to be substituted out. They're not working, so they're not learning at all.

So this is what I wanted to do that's different in this model than in all our other models. I want us to think about how technology doesn't affect us permanently. When you think about tractors: Tractors killed the farmer, and they killed the manager. We didn't need anyone to manage the workers in those, so it killed a whole line. What I'm thinking about with this technology that people are talking about is, when we're young, it might displace us, but when we're old, it kind of complements us. So it's really about career choice. If it "kills" you when you're young, you just become an "old" person quicker, exactly like we did earlier. That's the trade-off here. Now, you could layer on: Suppose there's another sector out there, and you could move to the other sector. Suppose there's a home sector out there, and you could move to the home sector. All of those things are going to be like what we did with every one of our models in terms of technological shocks and how it affects the labor market forever, so those could be layered on top. But I really wanted to think about the career-ladder part.

Audience question: It kills you when you're young because, presumably, the more automated things are, then that distance is smaller.

The distance is smaller, so you learn more. You learn less at that point, and that's the whole thing.

Audience question: Did you say tractors or trash stairs? I'm joking. {Laughs} Okay, I have a question. I guess what I'm stuck with is, I understand the model. But if only machines work, I thought the role of the manager was to organize workers and activities. Once they have just machines, the production function changes ...

That's not what managers do here. Managers come up with new tasks. That's what they're doing in our model. They're innovators. There's no manager in the production function here. We could put managers here, but that's not what they're doing here. They're just creating new tasks.

Okay, today I'm just going to shut down any human capital growth for managers in the equations going forward, and I already told you I'm not including Martin's effect on the worker side. The technology won't affect that Δ. I'm just going to subtract it.

We're going to do two versions of this model. I'm going to show you both in parallel as we go through. First, I'm going to show you the planner's problem of this model, where there's no inefficiency anywhere. The firm is going to internalize when they automate or when they expand the frontier, how much is going to affect the learning rates of workers. That's going to be version 1. I'm going to call that the planner's problem all the way through. The planner is going to maximize the total value of the economy, basically how much is produced. You have to subtract the part you pay for AI. You have to subtract the pay for the workers, and you have to subtract the costs you pay for the managers.

Audience question: I was going to ask you that — why is it that managers don't internalize ...

That's the planner's problem: Everyone internalizes everything; there's no inefficiency.

Audience question: What about in the equilibrium? Something is not going to be internalized?

In this version, everything is internalized. I'll do a second version of the model. It's going to be close, a second version of the model, where it's going to be the pessimistic version of the world. We're going to allow this to potentially occur, which is that firms are not going to internalize the effect of their decisions on the human capital component. Now, why would they not do that? Maybe they're short-sighted, who knows. I could write down a model where they post how much they're going to innovate and what wages they're going to do, and you give me a model with a lack of commitment. The firm's going to have an incentive to deviate at every one of those moments. In that type of world, the decisions are not going to be internalized.

I'll run both of these parallel to each other, and this is going to be something that I show you as we go along. The planner's problem and the decentralized problem overlap over most of the range. There's only a small part of the range where they differ, and they're going to differ by — I'm going to use the word small, that's a quantitative statement — they're going to differ by a small amount depending upon what I mean by small as we go through.

Audience question: I was curious if the tasks can also become smaller. Can you delete tasks as a manager?

As you're going through, they have the marginal product. They get "innovated away" so long as they're done by machines, but maybe they have the marginal product of something very close. That's going to be the game.

Okay, I'm going to try to break through the equilibrium. The model is simple in some respects, but there's going to be a lot of rich economics in it. I'll focus everything through the labor market of workers. There will be a few equilibrium conditions, and then I'll go through simple examples and then get more and more complex to show you how the economics of this model is going to work. There's going to be a lot of rich economics, but it could all be broken down into demand and supply of workers as we go through.

Just like in Acemoglu-Restrepo, there's an automation threshold, τ^*, in the background. You're going to automate tasks below that; you're going to use workers above it. This is going to be a function of how much you have to pay workers relative to the price of the machine. So the wage of workers, an endogenous variable, is embedded in the automation decision. We're going to have to keep track of that.

Okay, this is from the functional form of automation. We could have a degree of freedom to play with this and other things, but the key thing is when workers are expensive relative to machines, you're basically going to automate more.

There's also an optimal frontier-expansion choice, where you're going to come up with new tasks. When you come up with new tasks, you're going to get the benefit of coming up with a new task. What is the benefit of the new task? I'm going to get some additional output, but I have to pay for that output either with machines or with workers. There's diminishing marginal product as we go through. You can see the wage is going to be like a labor demand curve, it's going to come directly from that part of the equation. As workers become more expensive, all else equal, the benefit of innovating (if you have to use a worker) goes down.

There's also a marginal cost that comes from doing that. If I'm going to expand the frontier, I either pay for a manager or I pay for a machine, with some elasticity between the two. So the innovation decision is a function of the wages of the worker and eventually the wages of the manager.

There's a labor market equilibrium, saying that the demand for workers. When do I use workers, for all tasks between τ* and F are produced with workers? I need some workers, and, again, there's a downward-sloping labor demand curve due to diminishing marginal product.

Labor supply is all the workers that are out there in the human capital space. They start with 1. x* is going to be the threshold at which a worker changes to a manager. Then I'm going to sum over all the workers in the economy. So this is the labor market. If I did the managers, there's another equilibrium for managers' wages. That's going to be the optimal stopping decision of when to become a manager. The labor demand and labor supply of managers are going to have to equal each other. I'm going to set in a world to build intuition where I'm going to make everybody a worker in this model, and then we'll bring back managers later. So that's why I'm going to focus on the worker part of this model.

So I'm going to start — again to build intuition of how this is working — with a simple model where we have endogenous learning, just like we did before, an exogenous frontier. This is basically just Acemoglu-Restrepo with endogenous learning. I'm then going to go through and endogenize the frontier, but I'm still going to assume there are no managers used in production. There's no value of managers in this economy in this second example. I'm going to assume the frontier just moves with the price of technology. Then we'll do the full model where we endogenize the manager choice as well.

Let's start with this simple example where there's automation, exogenous learning, no career choice, no frontier expansion — just Acemoglu-Restrepo with learning-by-doing. We have the endogenous learning rate, φ, which is going to be Δ times the amount of stuff done by workers. I'm fixing f to f-bar exogenously, no career choice. I've already said that four times. I'm going to start with the decentralized model to build intuition, then we'll bring back the planner's problem. So I'm going to show you all the results, just the first few sets of equations, assuming the firm doesn't internalize their choice of automation on the learning process, and then we'll allow them to do that. I'll show you where it's similar and where it's different.

Okay, so let's break it down. I'm going to go through a little bit slower for the next three or four minutes to build intuition, and then I'll move a little quicker because everything will be very similar after this.

So, human capital dynamics. You're going to grow at ρ-φ by the human capital stock in the economy. This is the aggregate human capital stock. We're going to die off at rate χ. When we die off, we lose our human capital and are reborn at a human capital of 1. In steady state, where we're basically in a world where human capital growth is stationary, the steady-state human capital level is just going to be related to χ relative to χ – φ.

We have a labor-demand curve in the economy, downward-sloping because of the diminishing returns. φ – Δ is just manipulating our learning function. Our learning function is basically φ = Δ * (f – τ*). How much is our total labor demand in the economy? F – τ*. I can re-express that as φ / Δ, just playing around with that learning function. So now I have a labor demand curve in human capital space that's a function of the wage in the economy, which is a function of φ. The labor demand curve is also directly a function of φ, and I have a labor supply curve that's also a function of φ. So I have both the supply of workers and demand for workers in the economy as a function of the actual amount of learning in the economy.

In this model, just as in Acemoglu-Restrepo, the wage of workers is pinned down by the τ* automation threshold, so just substituting in the threshold for our endogenous τ* (which is basically just f – φ over Δ), just playing around again with that learning function. I have an equation now for the equilibrium wage of workers. So what I'm going to do is I'm going to play around with the labor demand curve in the human capital space, the labor supply curve in the human capital space, knowing that the equilibrium wage in the economy is going to be positively related to the amount of Q. That's the exogenous variable now. I'm going to shock Q. God's going to wake up and make this technology cheaper. That's the exogenous shock in the model. When the world gets cheaper, what happens? Wages of workers fall, just as in Acemoglu-Restrepo. If technology is cheaper, I can automate more stuff. I don't need workers as much, so the demand for workers is going to fall, lowering the wage of workers. So these are the key things we're going to play around with now. These two functions, and knowing that when Q falls, wages are going to fall in the economy through the first-order condition for automation.

So, what does this human capital demand function look like? If φ is zero, basically there's no learning in the economy, and there's no human capital in the economy. There's no demand for human capital in the economy, so that'll be zero. When φ is zero, we know the demand curve is going to be equal to zero. When φ is zero, supply is still going to be positive. In this case, it's going to be 1 χ / χ. As φ goes to χ, that's going to be the bounded equilibrium in the model. You get too much of that, just like in any growth model, you have explosive growth. So when φ approaches χ, what's going to happen there? Demand is going to be finite. Φ will be χ, and supply will be infinite.

So I have these two functions: one demand curve that's somewhat concave, and a supply curve that will be somewhat convex. As a result, this model is going to have potentially three or one equilibrium at any point in time. These curves are going to intersect twice, a low-learning and a high-learning equilibrium. And then there's always going to be an equilibrium where you could be at the frontier of learning. Basically you have max learning in this model. What is max learning? There's no automation whatsoever. F is equal to F-bar.

So what is this low-φ equilibrium? It's what I walked through in the introduction. There's going to be low labor supply, which is going to be high wages for every given unit of human capital. When workers are expensive in human capital space, they're going to automate more, which is going to mean less learning, which means low labor supply, etc. There's a high-human-capital version of that as well.

Audience question: Call it the φ-age — this one's unstable?

No, this one's not going to be stable. In three slides, I'm going to show you the stability of things. Just as Nico [Trachter] said, stable, stable, unstable.

So what happens when Q declines in this case? When Q declines, worker wages fall. When worker wages fall, you're basically going to have the labor demand curve shift up. Workers are going to be cheaper, all else equal, in human capital space. It's like you're moving up the labor-demand curve, in this case, holding φ fixed. In the low-learning environment, you're going to want to learn less; in the high-learning environment, unstable, you'd want to learn more, and then you're still bounded by the frontier.

So moving Q is just going to see whether these lines are far apart or near each other. If Q ever gets sufficiently too high, you'll be in a world down here where the only equilibrium is going to be the frontier. And that's what it's going to look like in steady state. We'll get to the stability and transition dynamics next.

So when Q is really high, there's no automation and there's full learning. Eventually, Q will get low enough, and as Q gets low enough, you start on this equilibrium branch over here. We'll call it the stable one, which you'll see in a couple seconds when I show you. You're actually going to be automating more and learning less. That's just Acemoglu-Restrepo.

The one other tweak in this model is that, because of learning, if you start high, I'm going to show you in a little while, you're going to stay — this branch is going to be stable. You're going to get less automation than you would get under the Acemoglu-Restrepo. So having endogenous learning is going to then delay — under some equilibrium branch — the rate at which you start automating things away. So even in Acemoglu-Restrepo, with the endogenous learning, you're going to have a range where you might delay the automation a little bit more than you would otherwise. Now in the end, when it gets cheap enough, you automate everything away.

Audience question: Sorry, quick question: You said one or three equilibria. I think it's yet to be seen whether this is a unique dynamic equilibrium or multiple? You could still have a unique dynamic equilibrium.

Let me show the next slide right here. Suppose I'm out of steady state. In this model, there's going to be a unique learning rate at every out-of-equilibrium, out-of-steady-state point in this model.

Suppose we're out of steady state. We are still going to have at any given moment that demand for human capital is going to equal the supply of human capital, and, at any given moment, supply of human capital is fixed. At any instant, it's fixed because you have to have learning accumulate over time. So I can invert this function and say there's some φ out there, at any given moment, that's a function of my human capital and Q. In this example that we just did, Q and φ are positively related to each other: When Q goes down, learning went down unambiguously.

So in this model, again, where I don't have the planner's problem — the planner's problem will have to be an additional term in this curve — you're going to have a differential equation at any given point in time that's going to determine the growth rate of our learning. It's going to be the evolution of the human capital process from the demand for human capital given that supply is fixed. And then supply is going to accumulate like we did before with learning at rate φ, given the human capital stock. We die off at rate χ, we lose our human capital stock and are reborn at 1.

So at any given moment, there's a unique equilibrium out of steady state that we're going to be moving toward a steady state when we go through it. I'm going to show you that exactly now — going back to Nico's question one more time. I could also do the stability of each of these steady states as well, just like we would do the stability of any of the other steady states in our model.

I thought the best way to show this is just via a picture. It's exactly what I showed before. The solid lines are steady states; the solid line is the stable steady state. The dashed line is the unstable steady state; the frontier is also a steady state.

So when you're way up here, there's only one steady state. When you're way down here, there's only one steady state. There are three steady states in the middle, two of which are stable, one which is unstable. At any given point in there, you're going to move toward the stable equilibrium, just like in any of our phases.

I'm drawing you these pictures in learning space, because that's what our paper is about, but you could also have this in human capital phase-diagram space. Your mind might want to be going to the human capital phase diagram, but I wanted to make sure the paper is really — I'm trying to think about what happens to learning with Q. So I've just inverted everything into the learning space.

If you start at the high equilibrium, you're on the frontier. If Q is high enough, we're at some frontier right now. We could stay there as Q goes down. Eventually, Q is going to fall sufficiently, and we're going to jump to the low-learning environment.

Last thing I'm going to do in this example — just to build some intuition — is show you the difference between the planner's problem and the decentralized equilibrium. They're nearly identical. There's a little range in between where the planner is going to want less

automation than the decentralized problem, because they're going to internalize the effect on learning. But all that really does is move us around a little bit in these kinds of intermediate ranges. They're going to want to stay at the frontier a bit longer before they jump. When they jump, they're going to jump pretty much to the same place. When technology gets cheap enough, even the planner is like, "Hey, why use workers? The machines could do it with all the benefit." So again, this is Martin's question: The difference between the planner and the decentralized is relatively small in this range. It need not be, but at the extremes they're identical.

Audience question: Did I miss anything? Can you start above the dotted line? And you would go to ...

You'd go to here, to the frontier. If you started way over here, you would go to the frontier at that point. There's no automation at this point, so you're going to get maximum learning the way it's set up. The frontier is exogenous — it's some number. If you're anything above that with no automation, you're going to go to that. So above, the phase diagram moves toward the top line. Anything below, you start moving toward the unstable low path.

Okay, so let's go to the one I think is richer. It's the opposite of Acemoglu-Restrepo: Suppose all the action is on the frontier. This is the model now: endogenous learning with automation and an endogenous frontier choice. We're going to do it simple in terms of the frontier choice, again to build some intuition. That simplicity is going to be: I'm going to assume we don't even need managers in this example, only the technology. The managers are going to dampen things, but the first-order effect comes from technology making the frontier cheaper.

So we're still going to come to our same labor-demand and labor-supply equations as we did before. The only difference in this example is that the wage is going to be pinned down by the automation, the frontier-choice decision between workers in the Q. Firms are going to choose how much to innovate. Q is going to affect the cost of innovation. Workers are going to affect the benefit. They're going to equate the two. What's going to happen then is the automation — τ^*— is going to be the residual that anything that the wage is given, the Q, the τ^* is going to absorb that. So in this model, all of the effect is going to be pinned down by the frontier.

Audience question: Let me give you the best shot for the pessimistic view to hold is the first simple model you showed us, where the learning, plus there's an additional effect, which is now the managers are also worse off. If we needed managers to create new tasks, that compounds. It's even worse.

So everything's giving the pessimistic view the best shot.

Audience question: This is the best — I guess example three. No, because example three is going to be the one where automation actually helps the managers. So that was the opposite direction.

Let me show you: Well, this one goes in the opposite direction too. So let me show you what it's going to do with this one as well.

Everything's going to be the same. The only thing that's going to be different between these two scenarios — example one and example two — is that when the price of technology goes down, the wage of workers goes up in this example. Why? Because when you're expanding the frontier as technology goes down, your demand for workers goes up because there's more stuff to produce, and that is going to raise the wage of workers. So in this example, a decline in Q raises the wage of workers. In Acemoglu-Restrepo, it lowers the wage of workers. That's the only difference between these two examples so far.

So, when you have this endogenous frontier — and we'll put it all together with the managers in example three. There is automation in this model. Automation is endogenous, but wages are going to be pinned down by the frontier choice, and the automation is just going to be the residual given the price and the wages that workers have.

You still have the same labor demand and labor supply curves in a human capital space. The only difference now is that as Q gets cheaper, the curve shifts in as opposed to shifting out, because workers are now more expensive. And as workers get more expensive, your demand, all else equal, is going to fall. So instead of having the inverse U-shape, you have a U-shape. As Q goes down, you again get three equilibriums: the stable-low, the unstable-high and then the frontier.

In this example, a lower Q is just going to increase human capital all the way through. Why? Because you're expanding the frontier. And as you're expanding the frontier, there's more to learn, and eventually learning is going to go up. If you happen to be at the frontier, you're going to stay at the frontier. If Q gets sufficiently low, the frontier goes out completely, and then we get maximum learning no matter what, because there's just going to be so many things for us to learn from.

So, to Martin's point, there's an intermediate step I'm doing between the two bounds that you did. When you endogenize the frontier, it's going to push a force in the opposite direction that is going to create learning opportunities, which is going to generate learning. And if you only have workers, this is the dominant force, because automation will just be the residual in the background, because the workers' wages are going to move around as the frontier is moving around. Now, it's the same Q in both the frontier and the automation, maybe they're different Qs. We have different instances like that, but in that world, unambiguously the frontier is going to be — the human capital is going to be growing as technology gets cheaper.

Audience question: Is this a theorem or a quantitative statement?

This is theorem so far. There's nothing quantitative in any of this.

Audience question: My question was kind of like: You had the two forces. One is the typical manager is fixing the number of tasks.

I have no managers in this; this is all workers. So the next one, that's why I said I'm trying to break out the manager part of all of this. So — Martin, this is what I'm going to use my last four minutes for.

I just wanted to say the planner is going to want more innovation, all else equal, so they're going to want more learning, all else equal, just like they did before. So that is going to have the same shape, just a little bit more convex than it did before.

Okay, last thing, and this I'm going to do very quickly, and then we'll go through. Let's put everything together. And this is not theorem, but let me tell you what forces are going to generate patterns like this.

There's going to be, as Q goes down, early on — and this is what Nico talked about earlier — this is going to be functional forms on the relative curvature of the task frontier relative to the curvature of the automation decision. In a world where it's Acemoglu-Restrepo and it's linear in the automation space, in any sort of complementarity, you're going to get a pattern looking like this. Early on, when technology falls, the marginal benefit of innovation is very high because of the complementarity between the workers. You get that extra kick going on, and it's linear in terms of automation.

So at high levels of Q, we start with no AI. AI comes in. You're going to get a range where you're going to be innovating if you are on the low-equilibrium sense, where you're going to be getting human capital, or you're already at the maximum and you're going to stay at the maximum. Eventually, Q is going to get sufficiently low that you're going to then jump to the complete-automation equilibrium. The planner wants that. We want that. This is the world where Martin said the robots are doing everything for us, where they're innovating for us and doing our job, that's a win-win for everybody.

So in this world — and this is Nico's comment early on — if the relative substitutability of the task frontier is more than the relative substitutability of the automation, you're going to get this kind of pattern in the data. That's a theorem that we kind of proved. But those functional forms, who knows what those functional forms are? But if you believe in Acemoglu-Restrepo and automation in some sort of complementarity in the innovation, you're going to get this.

So this is a pretty optimistic story that we start at infinity, it comes along, either we're in the low-learning environment and we're going to learn more, or we're already at the frontier and we're going to stay at the frontier. Then, if technology gets low enough, then we're

going to jump to the low-learning environment. But that's the "lottery ticket": robots doing everything for us.

And the planner's version is very similar to that as well. The planner wants a little more innovation, a little less automation, but the same thing is going to be that both are going to be increasing as if you're in the low equilibrium, you're going to stay on the high equilibrium, and then eventually even the planner is going to want to jump over to the low-learning environment, because the robots are doing everything.

So that's a more optimistic version. I did stack the deck to give the pessimistic version a shot, but once you give me an endogenous frontier, it pushes in the other direction. That pushing in the other direction could actually sustain human capital. So maybe the grad students coming through will still be productive tomorrow: You'll still have value in your jobs even as you get older.

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